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DSC640 – Data Presentation & Visualization
Professor Benjamin Schneider
November 2, 2025

Final Paper

The National Database of Childcare Prices maintains an exhaustive list of data that goes beyond the cost of childcare. While that might be the sole purpose for the data, one glimpse and we can find other branching topics, such as unemployment rate, stratified by age and gender. Each row corresponds to a time and a place, such as Autauga County, Alabama in 2009. The primary key will likely be the State, Count_Name, and StudyYear. The various columns provide critical information, with a mix of categorical codes and numerical data.

The volume of data is more than we need for a simple analysis, as the data contains 227 columns. However, 15 columns have the same values across all rows, so they can likely be dropped. If one were to use this data source for a study, they would have to decide whether they want to do a deep dive into a specific demographic, such as females between 20 and 64 with children under 6, or the general population. The large number of columns leave the possibility to pivot into different areas.

There appear to be various “target” columns, such as Median Price for Centers, Median Price for Family Child Care, 75th Percentile Price for Centers, 75th Percentile Price for Family Child Care. For our purposes, we’ll focus on Median Price for Centers. We will explore the rising costs overall and for each unique age group (such as birth to 5 months, 6 months to 11 months, etc.

There are also 2 forms of incomes/earnings, one using the target year and one using 2018 dollars. 2018 dollars take inflation into account, which provides a better picture of the economy, so we’ll use those columns. While the data provides columns for households comprising two

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races, the majority of datapoints fall under 1 race, such as black, white, Asian, Hispanic. We will use the 1 race columns for our efforts, should we pursue insights into racial demographics.

Scenario:

For our efforts, we will use this data as the source for a political movement. We are imagining that the year is 2028 and the presidential election is upon us, and the key issue is childcare costs. Both candidates are neck-and-neck in the polls, and we have a last campaign push to win the election. Our goal is to present the data with a compelling argument that we understand the cost of childcare and what the driving factors are. Here are the questions we aim to answer:

1. What is the average cost of childcare?
2. What percent of income does the average household spend on childcare?
3. Is the data evenly distributed or is it skewed (indicating imbalanced in childcare costs)?
4. Do childcare costs cause poverty/unemployment?

Because we need to choose 3 different mediums, we will choose 3 different audiences.

1. Video – This will be a campaign speech. The duration will be between 3-4 minutes. This will provide us the opportunity to describe the problem, present a solution, and show off our expertise.
    a. The audience will be the **audience at the campaign speech** and **YouTube viewers** who will likely watch only the 3–4-minute snippet.

2. PowerPoint – This will be an internal "factoid" PowerPoint that will be up on the campaign website. It will be a brief set of slides (3-5) that rephrase the campaign's position on childcare costs.
    a. The audience of the PowerPoint will be **campaign volunteers** who will learn the facts so they can present them
3. Single page PDF printout – This will contain an even more condensed version of the PowerPoint. This will be for infographic leaflets to be handed out as physical copies.
    a. The audience of the leaflets will be voters on the streets, door-to-door voters

Chart Creation and Clean Up

Since the preliminary analysis, much was changed in my assumptions. While the original goal is still in sight, my knowledge of the data has changed. The initial analysis was based upon the nature of the data by understanding each column and how it played into the central cost of childcare. After I understood the columns more, I preemptively made a goal believing the cost of childcare was high. My goals shifted when I discovered that wasn't necessarily the case. I've instead focused on telling the story of data, regardless of whether it aligned with my original goals.

- Summary of Analysis
    - Findings
        - When controlled on a national scale, the cost of childcare takes up 10-15% of the median income. The cost of childcare is not strongly correlated to

any individual feature, but it was slightly correlated to the poverty rate of families.

- Assumptions
  - I began the project assuming the cost of childcare is related to poverty and income rates. I also thought the cost of childcare was skyrocketing which was causing women to be unemployed. While childcare costs rise over time, they are not correlated to female unemployment rates or poverty rates. Again, this is when analyzing the data on the national scale.
- Items that still need clarification
  - I will refine the heatmap and correlations scores. I might also investigate weighted variables. Right now, I use each row as a datapoint, but this lets low population areas be equally weighted to higher population areas.

- Direction of Story/Plan of Attack/Message you want to get across
  - This changed after analyzing the data more. Instead of telling the same story 3 times, I will choose 3 different stories, one for each medium.
  - My 3 medium choices will have 3 different messages.
    - Campaign Bob's PowerPoint – This is the message of the status quo campaign. Keep the current policies in place because they're keeping the cost of childcare low.
    - Campaign PSY's PDF – This is a leaflet dropped off to voters in a door-to-door campaign. These visualizations demonstrate how the status quo is failing Americans with rising childcare costs.

        - PolitiFact TikTok – This is a short form vertical video that debunks some of the facts within the political narrative. It shows charts but then explains the conclusions.

- Target Audience
    - The audience is dependent on the medium.
        - PowerPoint – campaign staff
        - PDF – Gen X/Boomers, homeowners who will have the PDF dropped off in-person
        - TikTok – This will be for young voters such as Gen Z, Millennials
- Mediums You Included & Why
    - PowerPoint – PowerPoints are a powerful medium for delivering points in a slide-by-slide manner
    - PDF printouts – Common for handing out in person, mass distribution
    - TikTok – Short form videos are powerful for delivering content and have been utilized for reaching demographics, especially in politics
- Design Decisions
    - For my designs, I used Canva, a free web app. Without using a paid plan, Canva helped me design mockups for the data. I used Python and matplotlib for generating my data visualizations.
- Ethical Considerations
    - Background

        - I had an ethical conflict after I discovered that the data did not match my goals. I thought that childcare consumed 20-30% of the income but it was close to 10%. Rather than try and make the visualizations so they supported my initial claims, I pivoted my goals to align with the data.
        - Additionally, I had a heated debate with a fellow classmate about the scope of my project. Because I had chosen to use all the data (meaning all states), I was "watering" down many of the characteristics. While I agree with his premise, I wanted to represent the story for a presidential candidate. This means I need to keep the scope national.
    - What changes were made to the data?
        - There were too many columns to consider all data points in the final project. I dropped several columns and focused on making visualizations related to income and demographics (such as male vs female).
        - Converted all the household columns to percentages by dividing by the Number of Households
        - Grouped by years for time series
        - Created new features to compare childcare costs as a percentage of median income.
    - Are there any legal or regulatory guidelines for your data or project?
        - There was an accompanying PDF/guide related to the childcare dataset.
    - What risks could be created based on the transformations or how the visualizations are presented?

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        - For my project, I chose a political campaign. The store is told from different perspectives. There is a risk of misrepresenting data or failing to capture the full picture, leading to a bad choice in policy or electing someone who is good at convincing people they can solve their problems.
    - Did you make any assumptions in cleaning/transforming or when presenting the data? Did you filter any data without labeling or clearly identifying that the data was not included?
        - I did make assumptions about data points by counting them 1 to 1. I clearly labeled the data I was going to filter out and explained why in comments and in my project proposals.
    - How was your data sourced/verified for credibility?
        - Yes, the data comes from an official source. It has been peer reviewed on Kaggle.
    - Was your data acquired in an ethical way?
        - Yes, the data is acquired from the government so it is directly sourced.
    - How would you mitigate any of the ethical implications you have identified?
        - Peer reviews have helped show criticism of my assignment. David Hatchett brought up some great points when I posted my visualizations in Teams. David and I had a debate, which showed my sincere concerns with the data not matching what I believed but showing that I was still pressing on.
- Lessons Learned
    - What would you do differently next time?

        - As David pointed out in my posts, I could detect more distinct patterns if I simply used state-level data. Next time, I would likely use the state of Nebraska for my dataset instead of viewing the data from a national context.
    - What did you enjoy the most?
        - What I most enjoyed was being wrong. I began the project with assumptions that were quickly disproven and I led the new evidence drive my mission, not my beliefs or emotions.
        - Additionally, the decision to view data from a national perspective was met with criticism from my classmates. I appreciated their feedback and I felt like the debate opened my eyes to a different viewpoint.